

Perovskite Photovoltaics in June 2026: Defect passivation, composition and phase control

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Abstract

An evaluation of 623 publications indexed during this monthly cycle, with 167 examined in critical detail, reveals that perovskite photovoltaic research remains divided between unverified laboratory claims and corroborated physical mechanisms. Independent certification remains scarce, as only 14 studies provide accredited performance data. The verified efficiency ceiling reaches 33.6% in monolithic tandem architectures through dense self-assembled monolayer (SAM) contacts that suppress buried-interface halide migration, whereas single-junction devices attain 27.28% through multidentate Lewis-base chelation of undercoordinated lead cations. A pronounced area penalty emerges upon device scaling: although uncertified small-area coupons assert higher values, certified power conversion efficiency (PCE) falls to 26.8% at an aperture area of 655.2 cm² because localized phase impurities, non-uniform film crystallization, and series resistance penalize large-area charge collection. Operational stability reporting exhibits severe evidentiary deficits, as only 6 investigations document operational lifetime, with the longest sustained test achieving 3600 h, while only 5 adhere to standardized International Summit on Organic Photovoltaic Stability (ISOS) protocols. This reporting deficit extends across the literature: although 48.1% of abstracts state an efficiency, only 5.9% report independent certification, leaving most performance claims unshielded from sweep-rate hysteresis, spectral mismatch, and transient capacitive artifacts. Mechanistically, corroborated evidence converges on multidentate coordination and stereochemical configuration to elevate halide vacancy formation energies, distinguishing genuine deep trap remediation from transient field-effect screening. In contrast, claims of thermodynamic phase stabilization remain thin, as chemical additives primarily introduce kinetic barriers or alter coherent interfacial strain to delay conversion from the photoactive cubic phase to the non-photoactive hexagonal phase. At selective contacts, hole extraction depends on molecular ordering, catalytic chemisorption, and chemical buffering against interfacial deprotonation rather than macroscopic work-function tuning, which conceals severe nanoscale trap disorder. Under operational electrical stress, non-recoverable decay stems from coupled interfacial degradation, including electrode corrosion and fullerene photo-oligomerization, demonstrating that unstandardized steady-state characterization routinely conflates reversible ionic polarization with permanent defect remediation.

Keywords: perovskite photovoltaics; buried interface; defect passivation; wide-bandgap perovskite; tandem solar cells; operational stability

1. Introduction

Standard steady-state characterization cannot distinguish reversible ionic redistribution from irreversible chemical decomposition, leaving ambiguous whether reported passivation strategies eliminate recombination centres or merely establish transient screening fields. Resolving this ambiguity defines the central challenge in perovskite photovoltaics, where the research frontier has departed from raw power conversion efficiency (PCE) recorded on small-area coupons. Progress now requires reproducible independent verification, structural uniformity over scalable device areas, and operating lifetime that resists coupled thermal, environmental, and electrical stress.

A total of 623 indexed works met the scope gate in June 2026 and 167 were read closely. Across this evidence base, this review distinguishes what the month’s literature verified through corroborated physical measurement from what it asserted through uncalibrated fits, isolated champion cells, or non-standard aging protocols.

Five mechanistic threads track the findings that withstand that measurement uncertainty: non-radiative recombination and defect control, crystallisation pathways and phase stability, interfacial energetics and contact design, operational lifetime and failure pathways, and subcell design and tandem interconnection. Tracing these processes clarifies where microscopic defect passivation translates into authentic device endurance, and where nominal performance gains remain artifacts of measurement conditions.

2. The Certified Frontier in June 2026

Unstandardized laboratory current-voltage measurements cannot distinguish interfacial defect passivation from transient capacitive currents, sweep-rate hysteresis, spectral mismatch, or aperture misalignment. Only 14 of 167 closely read papers reported an independently certified efficiency, leaving the majority of performance claims unshielded from these experimental ambiguities. What survives standardization establishes the verified photovoltaic ceiling. Two-terminal (2T) tandems set the highest certified power conversion efficiency (PCE) at 33.6% and 32.95%, with a lower certified baseline at 28.79% [1,2,3]. Unclassified cell architectures reach certified efficiencies of 27.28% and 26.84% [4,5], while certified inverted (p-i-n) devices cap at 26.8% and 26.44% [6,7].

For scalable modules, the certified ceiling drops to 26.61% [8]. Self-reported values at large aperture areas run substantially ahead of this verified module boundary, reaching 27.12% on 655.2 cm² [6], 26.88% on 29.7 cm² [9], 26.76% on 12 cm² [10], 26.52% on 50 cm² [11], and 26.13% on 57.6 cm² [2]. A champion efficiency recorded on a tiny laboratory cell and an independently certified efficiency measured at module area provide fundamentally distinct physical evidence. The miniature laboratory cell interrogates local non-radiative recombination suppression under negligible series resistance and idealized collection geometries. By contrast, a certified module reflects distributed transport physics, penalizing performance through interconnect dead areas, sheet resistance across transparent conductors, and film thickness gradients over large spans. Treating micro-scale champion metrics as indicators of module feasibility conflates intrinsic material quality with macroscopic charge-transport management.

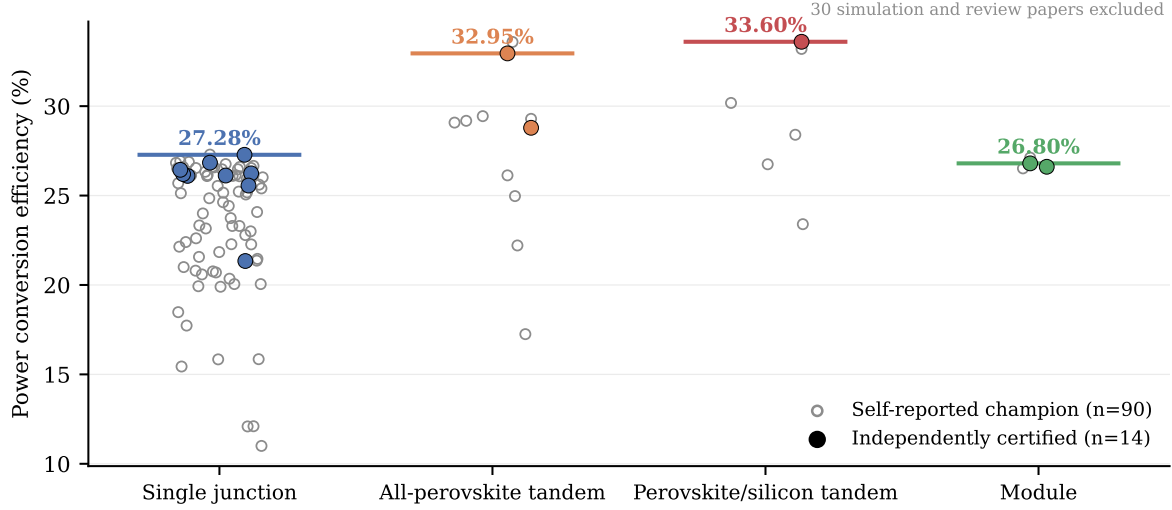


Figure 1. The certified frontier in June 2026. Filled markers are independently certified efficiencies, open markers are self-reported champion values, grouped by device family. Horizontal bars mark the highest certified value in each family.

3. Non-Radiative Recombination and Defect Control

Standard optoelectronic characterization cannot definitively separate true chemical passivation of deep sub-gap trap states from electrostatic band-bending, interfacial dipole reorientation, or suppressed mobile ion accumulation that dynamically alters interface recombination. Most reported open-circuit voltage (V_{oc}) and photoluminescence improvements conflate these effects. Evidence of genuine chemical trap remediation survives where multidentate Lewis-base chelators target undercoordinated lead (Pb^{2+}) cations and halide vacancies. Bidentate coordination directly elevates defect formation energies and prevents ligand detachment: dabrafenib (DRFN) binds multiple Pb^{2+} centers through its sulfonyl group and heterocyclic nitrogen to yield uncanceled power conversion efficiencies (power conversion efficiency (PCE)) of 25.28% in n-i-p and 26.62% in p-i-n cells [12], while dithiol chelation (DPS) forms stable five-membered rings with Pb^{2+} to achieve 22.28% PCE [13]. Enantiomeric anchoring reveals that steric configuration dictates coordination strength: chiral R-Pye raises the iodine vacancy formation energy from 2.46 to 4.56 eV with a 9.45 eV binding energy compared to 7.33 eV for S-Pye, delivering 26.13% PCE [14]. Similarly, functionalized aromatics like 2-amino-4,6-dichlorotriazine relieve PbI_6 octahedral distortion while coordinating Pb^{2+} and halide vacancies (26.58% PCE) [15], paralleled by triazine anchoring that suppresses Pb-I antisites [16], amino-pyridine derivatives (ACA) strengthening Pb^{2+} coordination via electron donation while hydrogen-bonding lattice iodide (15.84% PCE) [17], and dual-site binding through carbonyl, phenolic, or cyano groups [18,19,20].

This mechanistic picture is obscured when molecular treatments simultaneously alter bulk crystal growth, leaving unclear whether non-radiative recombination decreases from localized defect passivation or reduced grain boundary area. The dual-functional ionic liquid PIT decouples these actions: its anion coordinates Pb^{2+} and organic cations, whereas its cation segregates to the upper surface to retard crystallization and enlarge grains, achieving 25.94% PCE [21]. Co-passivation with potassium diformate and potassium phosphate similarly halves trap density while expanding grain size from 780 to 1370 nm (24.42% PCE) [22]. At buried contacts, separating chemical passivation from nucle-

ation control remains equally challenging. Adding CsPbBr₃ nanocrystal seeds passivates the buried interface while steering crystallization, yielding 24.63% PCE against 22.89% for unseeded controls [23]. Interfacial polymeric interlayers like ultraviolet (UV)-crosslinked RM82 diacrylates passivate buried tin oxide (SnO₂) interfaces while promoting crystal growth, though reported stability relies on passive ambient storage rather than operational stress [24]. CTW-g-PANI coordinates undercoordinated tin sites to heal SnO_x oxygen vacancies and passivate surface perovskite states (17.73% PCE) [25]. In tin-lead perovskites, differential chelation resolves mismatched crystallization kinetics: 1,10-phenanthroline binds Sn²⁺ preferentially over Pb²⁺, synchronizing film growth to achieve an uncertified 23.63% PCE, 81.27% fill factor (FF), and 0.887 V V_{oc} [26].

Static defect metrics also cannot distinguish permanent electronic trap deactivation from temporary trap filling or the suppression of field-driven ion migration. Although illumination-assisted trap filling achieves long electron lifetimes up to 547 ± 9 μ s in polycrystalline films [27], unpassivated mobile ions migrate under operational bias to generate new recombination centers. Defect suppression only survives long-term operation when ion migration pathways are physically sealed. 2PyS coordination suppresses undercoordinated Pb²⁺ and halide vacancies, limiting ion drift and preserving over 91% efficiency after 2000 h in wide-bandgap (WBG) cells [28], consistent with molecular confinement domains restricting formamidinium motion to arrest halide segregation (24.21% and 21.20% PCE for 1.68 and 1.78 eV bandgaps) [29]. Time-of-flight secondary ion mass spectrometry (ToF-SIMS) confirms that multidentate ATZCl forms a Ruddlesden-Popper (RP)-like surface phase that blocks iodide migration across the hole transport layer (hole-transport layer (HTL)) (25.22% PCE) [30]. Replacing small molecules with in situ polymerized vinylphosphonic acid provides a rare certified 26.24% PCE (26.54% uncertified champion) with over 90% retention after 1600 h of continuous maximum power point tracking (MPPT) [31], demonstrating that covalent network immobilization prevents field-induced passivator detachment and settles the ambiguity between chemical trap deactivation and dynamic ion blocking.

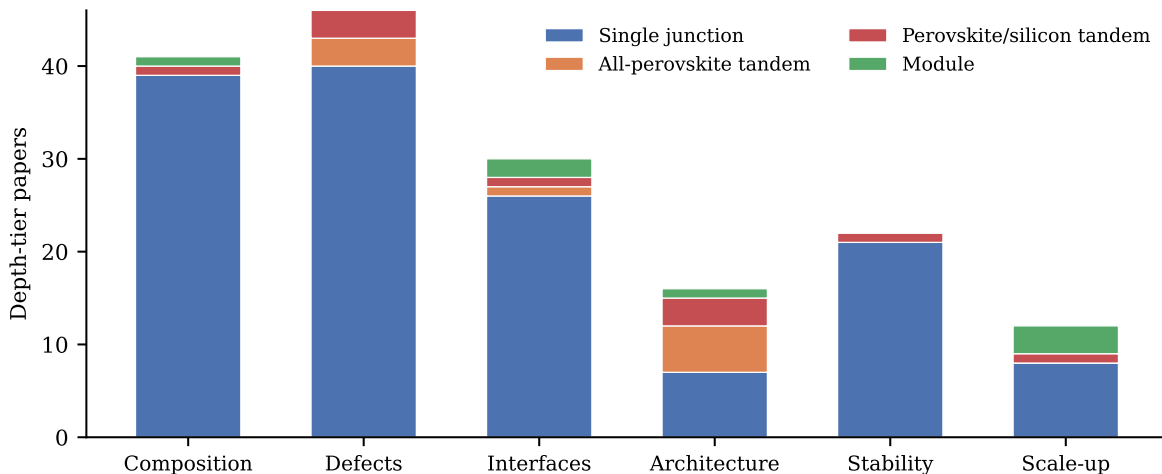


Figure 2. Where the month’s mechanistic effort was directed. Depth-tier papers by mechanism axis, stacked by device family.

4. Crystallisation Pathways and Phase Stability

Bulk structural probes cannot distinguish whether additives genuinely alter the thermodynamic free-energy landscape of perovskite phases or merely impose kinetic barriers that freeze metastable states [32]. Spatially averaged X-ray diffraction (XRD) routinely masks localized phase impurities, while thermal calibration offsets skew the determination of true phase-transition and decomposition onsets in halide powders [32,33]. What survives this diagnostic ambiguity is that local interfacial lattice matching and coherent strain govern polymorphic stability. In formamidinium lead triiodide (FAPbI₃), the spontaneous conversion from the photoactive black cubic perovskite (α -phase) to the non-photoactive yellow hexagonal (β -phase) at room temperature nucleates at coherent interfaces between α -phase (011) and β -phase (110) planes under tensile strain [34,35]. Introducing two-dimensional (2D) perovskitoid interlayers disrupts this preferred β -phase orientation, yielding an uncertified 25.61% champion power conversion efficiency (PCE) [34]. Interfacial strain regulation is equally achieved by constructing one-dimensional (1D) THAPbI₃ heterojunctions using tetrahexylammonium iodide (THAI), which anchors formamidinium (FA⁺) cations through hydrogen bonding to suppress outward cation diffusion and strain gradients, achieving an uncertified 25.06% PCE on unstated cell dimensions [36]. Transient intermediate phases also dictate phase-pure crystallization pathways. Volatile thiocyanate (SCN⁻) anions transiently coordinate with lead (Pb²⁺) precursors to restructure crystallization kinetics; while ammonium (NH₄⁺) cations volatilize during thermal annealing to drive bulk grain enlargement, residual phenylethylammonium (PEA⁺) cations segregate to grain boundaries to provide surface passivation, achieving 23.3% PCE [37]. In all-inorganic cesium lead triiodide (CsPbI₃), formamidinium acetate (FAAC) reduces the formation energy of the Cs₄PbI₆ intermediate phase, facilitating complete dimethylammonium (DMA⁺) to cesium (Cs⁺) cation exchange and accelerating dimethylammonium iodide (DMAI) removal to yield 21.84% PCE [38].

In wide-bandgap mixed-halide absorbers, measurements similarly struggle to distinguish growth-induced compositional heterogeneity from operational light-induced phase segregation [32,39]. Resolving this requires synchronizing disparate halide crystallization kinetics. Eutectic urea phosphate, possessing stronger affinity for bromine-based over iodine-based octahedra, equalizes reaction rates to eliminate vertical and lateral phase segregation, yielding 22.27% PCE [40]. Epitaxial growth on spinel oxide substrates uses lattice matching and magnesium-halide interfacial bonds to induce compressive strain, which arrests halide segregation without organic additives [41]. For multi-junction absorbers, substituting bromide with chloride in Cs_{0.05}(FA_{0.90}MA_{0.10})_{0.95}Pb(I_{0.95}Br_{0.05})₃ (incorporating methylammonium, MA⁺) circumvents performance losses observed beyond 450 nm thickness, enabling crack-free vertical crystallization up to ~700 nm without sacrificing fill factor (FF) or open-circuit voltage (Voc) to achieve 23.4% PCE on a 1 cm² aperture [42].

Controlling orientation templates is critical for dimensional stability. Templated crystallization using pre-synthesized large-area $n = 1$ 2D perovskite nanosheets drives three-dimensional (3D) film growth with high crystallinity and preferred orientation, achieving 21.36% PCE (20.75% stabilized power output) and a hysteresis coefficient below 1% [43]. Propylammonium iodide (PAI) added during two-step blade coating directs (111)-oriented growth in inverted cells, delivering 20.05% PCE and retaining 95% of initial efficiency over 1680 h of maximum power point tracking (MPPT) [44]. Conversely, solvent-additive approaches such as rubidium chloride (RbCl) in 2-methoxyethanol and dimethylacetamide (DMAc) claim methylammonium lead triiodide (MAPbI₃) phase regulation, yet stability verification is limited to 300 s of steady-state output at 18.48% PCE [45]. In quasi-2D

systems, functional group configuration in isomeric additives dictates molecular stacking: cytosine exhibits stronger delocalized interactions than iso-cytosine, facilitating favorable nucleation to attain 22.4% PCE with a T80 lifetime (time to 80% initial efficiency) of 3600 h [46]. Intermolecular pi-pi interactions in 3,3'-methylenediphenyldiammonium cations stabilize a non-Dion-Jacobson (non-DJ) 2D perovskite, yielding 26.52% module PCE on 50 cm² [11]. For tin-based perovskites, density functional theory (DFT) guided screening of solvent vapor fuming identifies tetramethylurea (TMU), which slows coordination kinetics to enable carrier extraction in 0.67 s and a 75.14 s recombination lifetime, reaching 10.45% PCE [47].

5. Interfacial Energetics and Contact Design

Macroscopic junction characterizations fail to distinguish whether contact modifications enhance carrier collection through electrostatic band alignment or through the physical elimination of local pinholes, solvent washouts, and interfacial deprotonation. Transient photovoltage and photocurrent mapping reveals that inhomogeneous hole transport layer (hole-transport layer (HTL)) junctions generate trapped-hole densities up to $1.6 \times 10^{17} \text{ cm}^{-3}$ [48], confirming that macroscopic work function shifts routinely conceal severe nanoscale contact disorder. Under operational stress, unimolecular self-assembled monolayer (SAM) contacts undergo severe degradation driven by thermomechanical mismatch with the perovskite absorber [49]. What survives this diagnostic ambiguity is direct evidence that contact durability depends on molecular ordering, multi-site anchoring, and chemical buffering against proton transfer rather than energetic tuning alone.

Because casting perovskite precursor solutions washes away unanchored molecules, contact integrity requires accelerated chemisorption or continuous replenishing. Catalytic deposition using 4-dimethylaminopyridine (DMAP) accelerates phosphonic acid binding to indium tin oxide (ITO) via phosphorylpyridinium intermediates, achieving a certified power conversion efficiency (PCE) of 26.44% (champion 26.93%) [7], while directly doping SAM molecules into precursor solutions or inserting alumina nanoparticles counters solvent erosion and seals pinholes [50,51]. At the substrate interface, uniform coverage requires suppressing molecular agglomeration; bisphosphonic anchors, seed layers, and rigidified conjugated backbones enforce packing uniformity and favorable dipole alignment without excessive islanding [52,53,54,55]. Specifically, rigid phenyl linkers elevate champion PCE to 23.30% compared to 22.38% for flexible alkyl analogues [54], and nonplanar butterfly geometries maintain 97.5% efficiency after 1000 h under International Summit on Organic Photovoltaic Stability (ISOS) ISOS-L-1 testing [55]. To resist interfacial shearing, multidentate and polymeric architectures buffer stress: a multidentate phosphonic acid network boosts fracture toughness nearly eightfold to 9.11 MPa (certified 26.61% PCE, though on an unscaled 0.06734 cm² area) [8], whereas polymeric Poly-3Ph-4PACz delivers 22.22% PCE on 57.6 cm² flexible modules [2].

At electron transport layer (electron-transport layer (ETL)) contacts, performance gains are frequently misattributed to work function realignment when chemical degradation dominates. Basic hydroxyls on tin oxide (SnO₂) promote formamidinium (FA⁺) deprotonation, a failure mode arrested by phosphate buffering or hydrogen-bonding additives at buried junctions [56,57]. Concurrently, bilateral functional molecules mitigate interface trap states by coordinating uncoordinated lead ions while covalently anchoring oxide hydroxyls [58,59]. Uncertified champion devices with L-isoleucine modification reach 26.15% PCE versus 23.54% for controls [58], while sulfonated bathocuproine derivatives suppress recombination through combined sodium-ion and pyridinic coordination to reach 26.02% PCE [59]. Decoupling bulk crystallization from interfacial reconstruction via triple-alkali interlayers

provides an open-circuit voltage (V_{oc}) of 1.184 V and 83.81% fill factor (champion 26.13% PCE) [60], demonstrating that chemical restructuring of the contact outweighs electrostatic band alignment.

6. Operational Lifetime and Failure Pathways

Standard operational metrics fail to distinguish reversible ionic polarization from catastrophic degradation under electrical stress, where cells with poly(triarylamine) (PTAA) suffer irreversible breakdown under reverse current near maximum power point (MPP) while carbazole-based monolayers show recoverable decay [61]. This diagnostic ambiguity is exacerbated by unstandardized testing: only 6 of 167 closely read studies reported a time to 80% initial efficiency (T80), and only 5 specified International Summit on Organic Photovoltaic Stability (ISOS) protocols [4,62,63]. Unmonitored aging conceals whether decay originates from intrinsic soft lattice dynamics [64] or external weathering, where outdoor testing of a triple-junction cell showed efficiency losses to 80% at 4 months and 50% at 6 months [65]. Accelerated stress reproducing 20 months of outdoor operation demonstrates that varying light intensity and bias drives three coexisting, spatially non-uniform degradation modes: copper corrosion, edge patterns, and phase segregation [66].

What survives this ambiguity is the mechanistic identification of non-recoverable interfacial failure. At electron-transport interfaces, phenyl-C61-butyric acid methyl ester (PCBM) undergoes photo-induced [2+2] cycloaddition, causing dimerization and morphological instability that steric additives suppress in uncertified 26.6% efficiency devices [67]. At hole-transport interfaces, chemical degradation dominates: unanchored lithium ions migrate unless coordinated by sulfur centers, which preserve approximately 90% of initial efficiency across 800 h of MPP tracking (maximum power point tracking (MPPT)) under thermal and light stress, albeit on 0.04 cm^2 areas [68]. Concurrently, self-assembled monolayer (SAM) destabilization is prevented either by transient two-dimensional (2D) capping layers that scale to 22.34% efficiency across 615.7 cm^2 modules or by hydroxylated MXene additives that strengthen SAM chemisorption on 0.0524 cm^2 cells [69,70].

Beyond interfaces, degradation is governed by deep-level rather than shallow defects: interstitial lead and formamidinium antisites dominate non-radiative losses despite concentrations three orders of magnitude lower than shallow states [71]. When exposed to operational heat, bulky surface passivators induce localized lattice expansion [4], whereas strain-regulating additives enable unencapsulated devices to retain 80.5% efficiency after 1000 h at 85°C under ISOS-D-2I and over 94.36% after 1500 h of MPPT under ISOS-L-1I [10].

7. Subcell Design and Tandem Interconnection

Macroscopic current-voltage measurements cannot distinguish benign structural disorder from performance-limiting defects: five sequentially evaporated scaffolds yielded identical single-junction power conversion efficiency (PCE) of about 19% (supporting an uncertified 28.4% champion tandem PCE) despite marked disparities in conversion, halide distribution, and morphology [72]. Nor can standard diagnostics separate electronic losses from mechanical fatigue, as bending stress in flexible all-perovskite tandems (uncertified champion PCE of 24.97%) accelerates non-radiative recombination detectable by grazing-incidence X-ray diffraction (GIXRD) well before crack formation [73]. What survives this ambiguity is that targeted chemical passivation and redox regulation reliably mitigate recombination and ion migration across buried contacts [1,3,9]. Anchoring dense self-assembled monolayer (SAM) coatings on reactive plasma deposition (RPD)-grown tin oxide (SnOx) suppresses halide migration in indium-free perovskite/silicon tandems, securing a certified PCE of 33.6% (uncer-

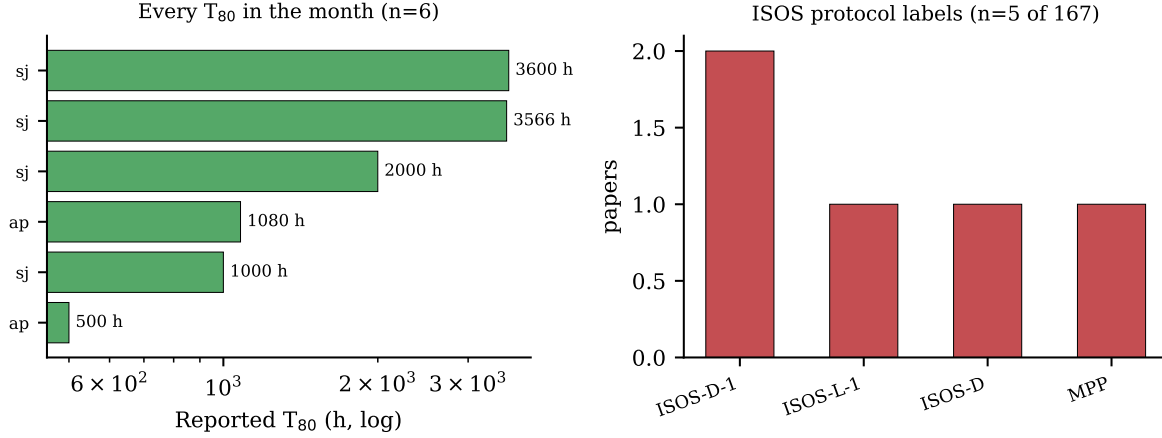


Figure 3. The month’s operational-stability evidence in full. Left, every reported time to 80% of initial performance. Right, the stability protocol labels used.

tified champion 33.2% on 1 cm²) and certified 31.0% on a 207.9 cm² minimodule [1]. At hole contacts, neutralized alkali metal-based phosphonate salts (2PACz-M) mixed with Me-4PACz delocalize interfacial charge across 29.7 cm² modules (uncertified champion 26.88% PCE) [9], while embedding allylthiourea (ATU) into poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate) (PEDOT:PSS) enables dynamic iodine recycling, yielding a certified 28.79% PCE (uncertified champion 29.29%) [3]. In narrow-bandgap subcells, Lewis coordination and seeding templates converge to govern crystallite growth: Lewis hard-basicity amplification achieved an uncertified 23.71% single-junction PCE and an uncertified 29.44% champion tandem PCE with an uncertified time to 80% initial efficiency (T₈₀) of 1080 h [74], while hybrid CsPbBr₃ quantum dot and metal-organic framework templates raise fill factor (FF) above 81% (uncertified 23.32% single-junction PCE; uncertified 29.18% tandem PCE, uncertified T₈₀ of 500 h) [75]. For scale-up, these chemical formulations endure atmospheric and mechanical stress: blade coating with low-hygroscopic alcohol-alkane mixtures at 80% relative humidity achieves a certified 26.1% PCE [76], while dynamic-bond cross-linking monomers permit self-curing at 40°C for 30 min to deliver a certified 26.8% PCE (uncertified champion 27.12%) on 655.2 cm² flexible modules [6].

8. Research Gaps and Outlook

Standard characterization cannot separate electronic defect remediation from transient ionic polarization or kinetic phase trapping. Certified power conversion efficiency (PCE) reached 33.6% in two-terminal (2T) tandems [1] and 26.8% across a 655.2 cm² aperture [6], yet only 14 of 167 closely read papers reported certified values, leaving most performance claims unresolved against measurement artifacts.

First, standard optoelectronic characterization fails to establish whether Lewis-base chelators permanently remediate deep sub-gap traps or merely induce transient field-effect screening [12]. Differentiating true chemical de-trapping from dynamic ionic screening requires operando, bias-dependent drive-level capacitance profiling (DLCP) across hole transport layer (hole-transport layer (HTL)) junctions, where interfacial trapped-hole densities reach $1.6 \times 10^{17} \text{ cm}^{-3}$ [48].

Second, bulk structural probes do not establish whether strain-regulating additives shift the equi-

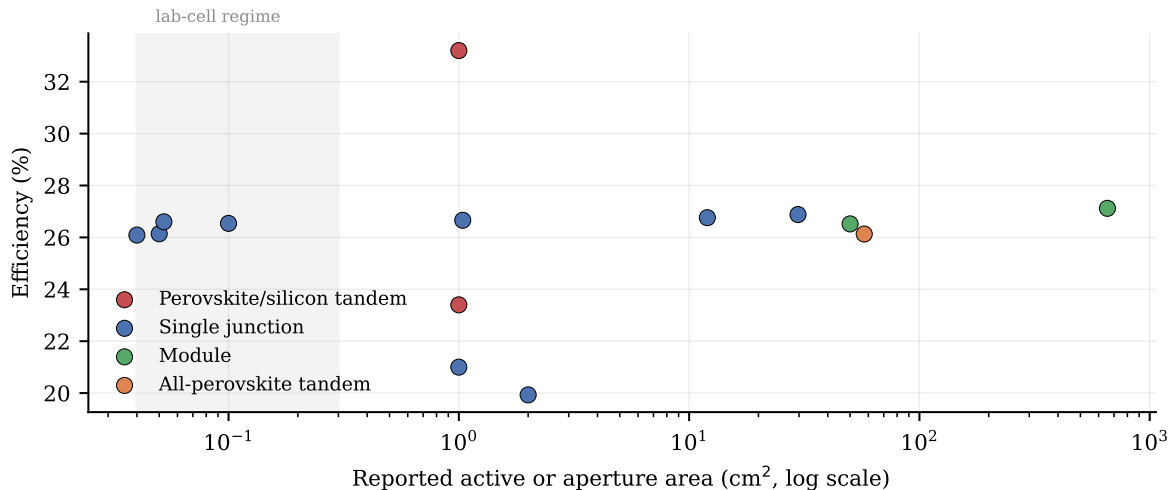


Figure 4. Efficiency against reported active or aperture area. The shaded band marks the sub-0.3 cm² laboratory-cell regime in which most champion values are measured.

librium free-energy landscape or merely impose kinetic barriers that freeze metastable phases [32]. Proving whether interfacial coherent strain suppresses yellow non-perovskite phase nucleation requires in situ high-pressure synchrotron X-ray diffraction (XRD) paired with differential scanning calorimetry calibrated against absolute transition enthalpies [34].

Third, unstandardized operational metrics fail to establish the boundary separating reversible ionic polarization from catastrophic degradation under electrical stress [61]. Only 6 papers reported a time to 80% initial efficiency (T80) lifetime (longest 3600 h) and only 5 papers named an International Summit on Organic Photovoltaic Stability (ISOS) protocol [4,66]. Closing this gap demands operando impedance spectroscopy tracking equivalent-circuit transport parameters during cyclic ISOS reverse-bias stress [4,61].

Fourth, macroscopic current-voltage scans cannot establish which morphological disparities in sequentially deposited scaffolds penalize carrier collection [72]. Although scale-up was addressed by 24 of 623 works, identifying performance-limiting defects over large areas requires spatially resolved photoconductive atomic force microscopy correlated with local photoluminescence quantum yields across device apertures up to 655.2 cm² [6].

Next month’s literature would be far more comparable if authors reported certified steady-state power tracking on standardized aperture masks alongside accredited T80 lifetimes under designated ISOS stress protocols.

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Declaration of Competing Interest

The authors declare no conflict of interest.

Data Availability Statement

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